

HAT-P-27b

R-0008

Benchmark run · workflow W8-benchmark · record 2026-bench_HATP-27_260707 · created 2026-07-08T03:46:45+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461228.7143 +/- 0.009 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.041
Transit depth (Rp/Rs)^2	1.92 +/- 1.13 [%]
Orbital Inclination (inc)	89.16 +/- 2.88
Airmass coefficient 1 (a1)	0.78 +/- 0.41
Airmass coefficient 2 (a2)	0.25 +/- 0.42
Scatter in the residuals of the lightcurve fit is	57.29 %
Best Comparison Star	None
Optimal Aperture	2.68
Optimal Annulus	7.0
Transit Duration (day)	0.093 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.139 ± 0.041 vs 0.119 (deviation 0.49σ)
Duration fit vs published	0.093 d vs 0.07 d (deviation 1.15σ)
Mid-time O–C	19.3 min
Depth SNR	3.4
Residual scatter	57.29 %

Light curve

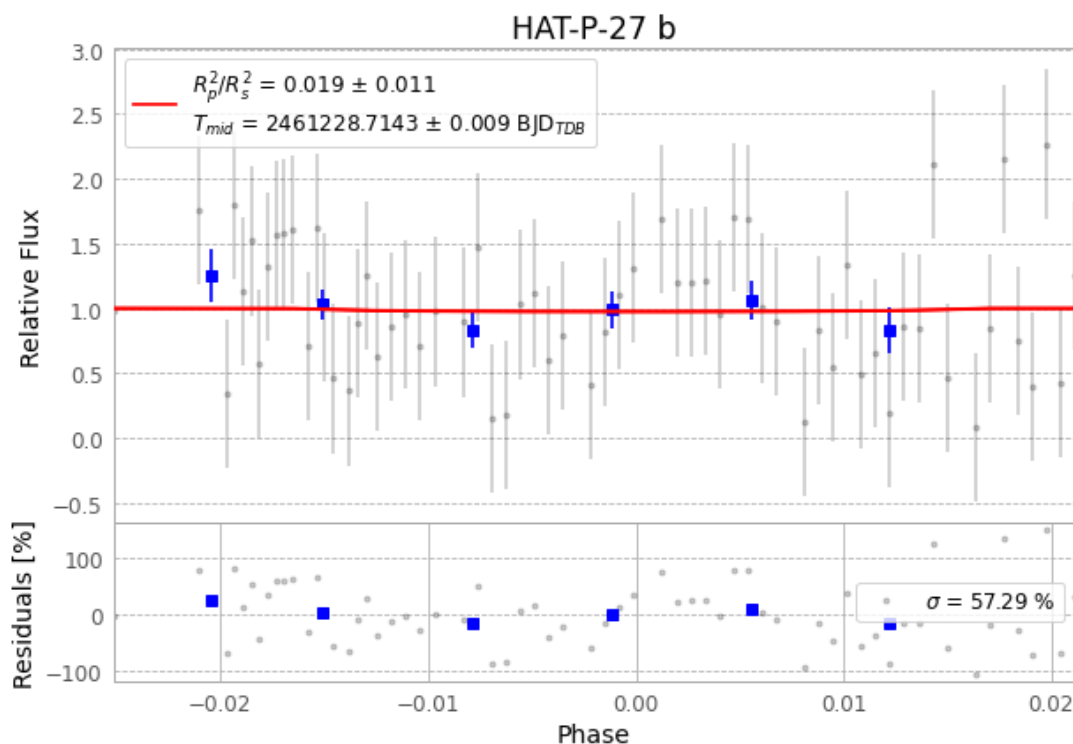


Figure 1 — FinalLightCurve_HAT-P-27 b_2026-07-06.png (SHA-256 6511e2efc0a4745e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [560, 146], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bc28102145e9050d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

75 FITS frames (49 MB), HATP-27260707025559.FITS → HATP-27260707063616.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-27-260707.log (538d270d70f8...), FinalLightCurve_HAT-P-27 b_2026-07-06.png (6511e2efc0a4...), FinalLightCurve_HAT-P-27 b_2026-07-06.csv (3340a7285d31...)

Compute resources

Wall time 173.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.